PCBC H Parametrics Practice Test D Name _____ _____/12

Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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_____

Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: the calculator question comes first, as announced in class. Every question copies a homework problem with new numbers. 3 points each is a practice split of the posted 12.

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth. *(practice instructions)*

1. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet.

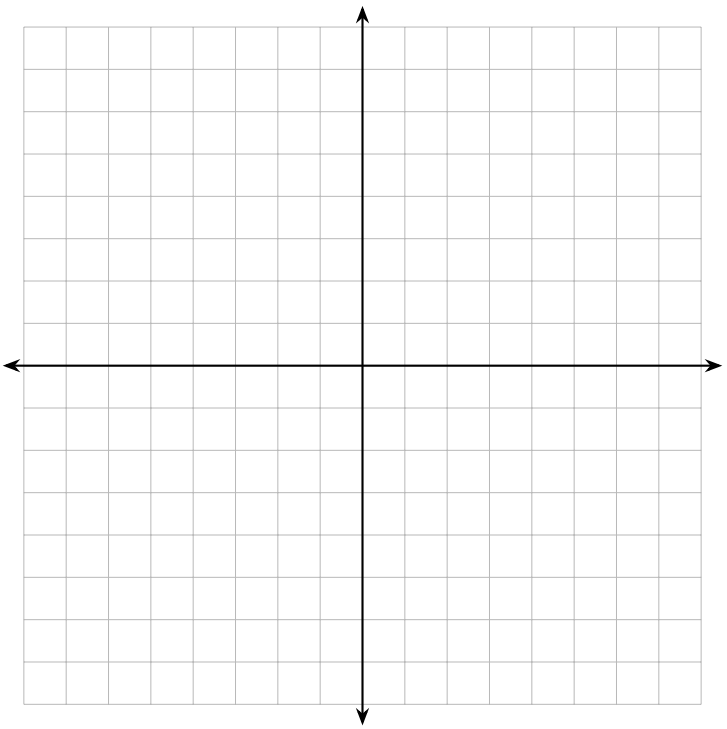
An archer shoots an arrow from a height of 4 ft at an angle of inclination of 25° with a velocity of 200 ft/sec. (a) Write the parametric equations for the arrow's path. (b) For how many seconds is the arrow in flight? What horizontal distance does the arrow travel? **3 pts**

Non-Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

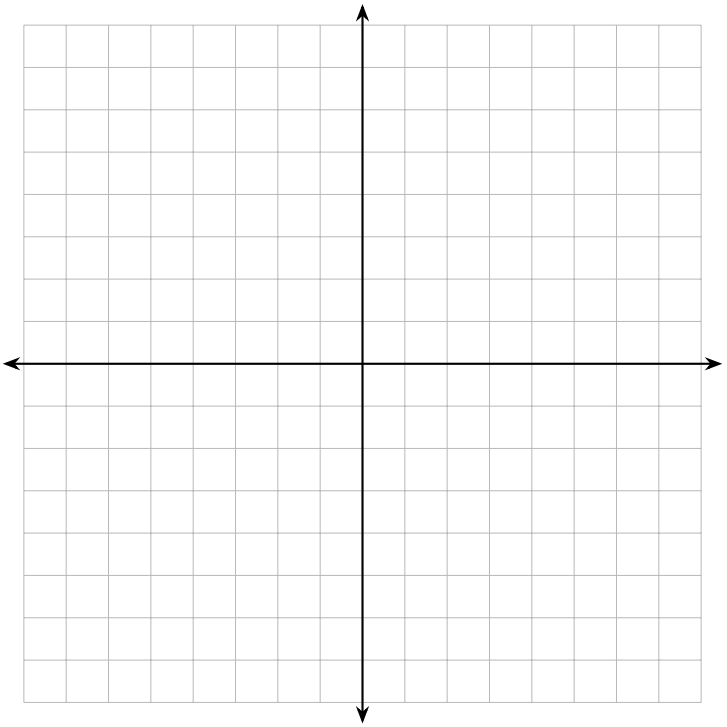
Find a rectangular equation for the plane curve, and graph the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

2. $x = t^2 + 2, \quad y = t^2 - 5, \quad$ for t in $(-\infty, \infty)$ **3 pts**



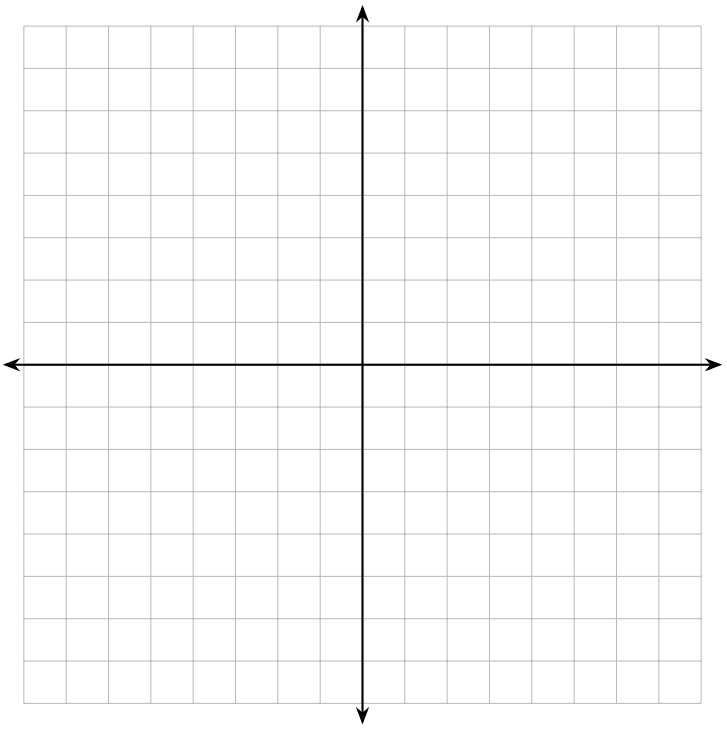
Find a rectangular equation for the plane curve, and graph the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

3. $x = 2t, \quad y = \sqrt{t^2+4}, \quad$ for t in $(-\infty, \infty)$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

4. $x = 3\cos t, \quad y = 5\sin t, \quad 0 \le t \le 2\pi$ **3 pts**



PCBC H Parametrics Practice Test D: Answer Key

12 points

Each solution follows the method used in class; the gray box shows another way. If the question does not say *simplify*, any form with t gone is fine (a form that shows the graph is best). *Simplify* means get y alone. Direction arrows only when the question asks. The calculator question allows small rounding differences. Every value was checked in Python.

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1. *Calculator* (like homework day 2 #20)

(a) $\boxed{x = (200\cos 25^\circ)t, \quad y = -16t^2 + (200\sin 25^\circ)t + 4}$

(b) in flight $\boxed{\approx 5.330 \text{ sec}}$, travels $\boxed{\approx 966.058 \text{ ft}}$

In class	Why
$x = (200\cos 25^\circ)t$	$v_0 \cos\theta$ times t, with the degree symbol.
$y = -16t^2 + (200\sin 25^\circ)t + 4$	Starting height on the end.
Degree mode. Parametric: $x1(t) = 200\cos(25)t$, $y1(t) = -16t^2 + 200\sin(25)t + 4$	Enter it exactly as written.
t from 0 to 6	The default 2π works here; raise it if the path stops early.
Trace to where $y = 0$; set the trace step to 0.01, then 0.001, until y is as close to 0 as you can get	Precision to the thousandths.
Read $t \approx 5.330$ and $x \approx 966.058$	Time, then distance, at the same point.

Other way. Quadratic formula on $0 = -16t^2 + (200\sin 25^\circ)t + 4$: $t \approx 5.330$ (reject -0.047). Then $x = (200\cos 25^\circ)(5.32964\ldots) \approx 966.058$ with the unrounded t.

Practice scoring (3 pts): 1 for the equations, 1 for the time, 1 for the distance.

Trap: radian mode, or using $t = 5.33$ to find x (≈ 966.124).

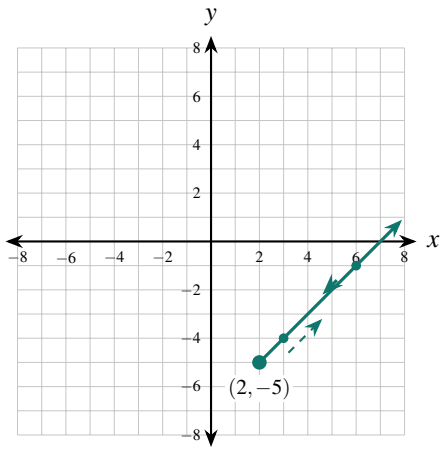
2. $x = t^2 + 2, \quad y = t^2 - 5, \quad$ for t in $(-\infty, \infty)$ *Non-Calculator* (like homework day 1 #5)

$\boxed{y = x - 7, \text{ for } x \text{ in } [2, \infty)}$
A ray starting at $(2, -5)$, solid dot there, arrowhead at the other end. It comes down to $(2, -5)$ and goes back up the same path.

In class	**Why**
t-chart: $t = -2, -1, 0, 1, 2$	Pick values on both sides of 0.
Plot: same points twice	t and $-t$ give the same point: it retraces.
$t^2 = x - 2$	Get t^2 alone instead of t (no $\pm$ needed).
$y = (x-2) - 5 = x - 7$	Substitute t^2 into y.
for x in $[2, \infty)$	Smallest x in the chart is 2; $t^2 \geq 0$, so x never goes below 2.

t	(x, y)
-2	$(6, -1)$
-1	$(3, -4)$
0	$(2, -5)$
1	$(3, -4)$
2	$(6, -1)$

Arrows: down toward $(2, -5)$, then back up. Only half of the line $y = x - 7$.



Other way. Subtract the equations: $x - y = (t^2 + 2) - (t^2 - 5) = 7$, so $y = x - 7$ in one line. The homework answer sheet draws the whole line; the t-chart shows x never goes below 2, so restrict it.

Practice scoring (3 pts): 1 for $y = x - 7$, 1 for x in $[2, \infty)$, 1 for the ray with a solid dot at $(2, -5)$ and arrows.

Trap: drawing the whole line. $x = t^2 + 2$ can't be less than 2.

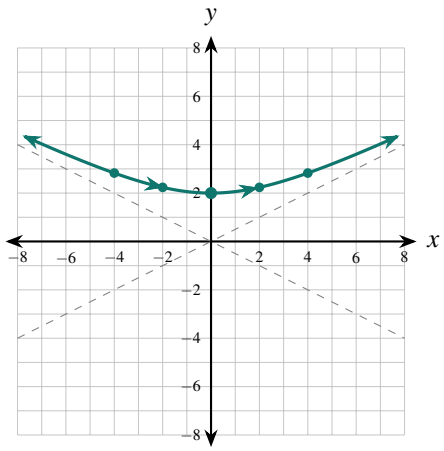
3. $x = 2t, \quad y = \sqrt{t^2+4}, \quad$ for t in $(-\infty, \infty)$ *Non-Calculator* (like homework day 2 #9)

$\boxed{y = \sqrt{\left(\frac{x}{2}\right)^2 + 4}}$ or $y = \sqrt{\frac{x^2}{4} + 4}$

x in $(-\infty, \infty)$. A hyperbola-shaped curve, lowest point $(0,2)$, traced left to right, arrowheads on both ends.

In class	**Why**
t-chart: $t = -2, -1, 0, 1, 2$	Parentheses with ∞: pick any t.
Plot, connect, arrows left to right	Lowest t to highest t.
$t = \frac{x}{2}$	Get t alone from x: one answer.
$y = \sqrt{\left(\frac{x}{2}\right)^2 + 4} = \sqrt{\frac{x^2}{4} + 4}$	Substitute into y.
x in $(-\infty, \infty)$	$x = 2t$ can be any number, so nothing to restrict.

t	(x, y)
-2	$(-4, 2\sqrt{2})$
-1	$(-2, \sqrt{5})$
0	$(0, 2)$
1	$(2, \sqrt{5})$
2	$(4, 2\sqrt{2})$



$\sqrt{5} \approx 2.24$, $2\sqrt{2} \approx 2.83$. A square root is never negative: top branch only.

Other way. Square both sides to see the shape: $y^2 = \frac{x^2}{4} + 4$, so $\frac{y^2}{4} - \frac{x^2}{16} = 1$, the top branch of a hyperbola with asymptotes $y = \pm\frac{x}{2}$. Keep the $\sqrt{\ }$ form as the answer.

Practice scoring (3 pts): 1 for $y = \sqrt{x^2/4 + 4}$, 1 for the top branch only, 1 for the graph with left-to-right arrows.

Trap: writing $y = \pm\sqrt{\cdot}$ or drawing a bottom branch. The original y is a square root.

4. $x = 3\cos t, \quad y = 5\sin t, \quad 0 \le t \le 2\pi$ *Non-Calculator* (like homework day 1 #6, #13 and #14)

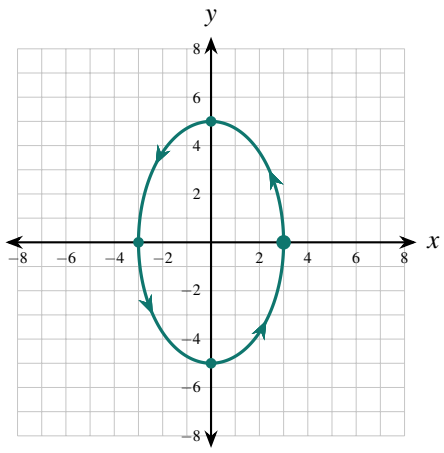
$\boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$

Whole ellipse, counterclockwise, starting and ending at $(3,0)$. No domain needed.

In class	Why
t-chart: $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$	Quadrant values.
Plot, connect, arrows: $(3,0)$ to $(0,5)$	Right, then top: counterclockwise.
$\cos t = \frac{x}{3}, \quad \sin t = \frac{y}{5}$	Isolate the trig.
$\frac{x^2}{9} + \frac{y^2}{25} = 1$	Square, $\cos^2 t + \sin^2 t = 1$.

t	(x,y)
0	$(3,0)$
$\frac{\pi}{2}$	$(0,5)$
π	$(-3,0)$
$\frac{3\pi}{2}$	$(0,-5)$
2π	$(3,0)$

Closed curve: it restricts itself.



Other way. Read it off: 3 in front of cos means 3 left and right, 5 in front of sin means 5 up and down, center $(0,0)$.

Practice scoring (3 pts): 1 for the equation, 1 for the ellipse drawn to size, 1 for counterclockwise arrows.

Trap: swapping the 9 and 25. The number with x goes under x^2.